# Andrea Muolo

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## EXPERIENCE

### Senior .NET Developer and Business Analyst
**Systemorph**

Dec 2022 - present | Zürich, Switzerland

Developer of the Systemorph Cloud platform and client-tailored web-apps.

Tech advocate, led POCs and responsible for meeting clients, absorbing business specifications, and implementation, from data modelling, to business logic (actuarial methodologies, accounting standards, etc.), business processes, UX design, and modern cloud architecture principles.

Organize and conduct presentations to clients, on-site trainings, and workshops.

Developer and tester of in-house C# modules, creation of concept applications to push the limits of our technology and foster our community of devs.

### Junior .NET Developer and Business Analyst
**Systemorph**

Dec 2021 - Nov 2022 | Zürich, Switzerland

Receive business specifications from clients and stakeholders and implement tailored data models, business formulas and processes in modular, scalable, and resilient web applications. Experience with actuarial methodologies and accounting standards. Implementations realized in the .NET framework.

### Postdoctoral Research Fellow
**Hebrew University of Jerusalem**

Oct 2019 - Nov 2021 | Jerusalem, Israel

Exact Factorization techniques in molecular physics. Algebraic manipulation of exact factorization equation and numerical calculations to tackle non-adiabatic systems. Genuine predictions on the Berry phase. Methods developed in C++20.

### Research and Teaching Assistant
**ETH Zürich**

Dec 2018 - Sept 2019 | Zürich, Switzerland

Many-body Dirac Equation and relativistic quantum mechanics. Novel analytical and numerical solutions to integrals to enable symmetry projected optimization.

Method development: Pre-Born–Oppenheimer and Tensor Networks algorithm.

Rich experience in concurrent and distributed computing using C++17.

## EDUCATION

### Doctor of Science Degree
**ETH Zürich**

Nov 2014 – Dec 2018 | Zürich, Switzerland

Explicitly Correlated Gaussians and the Quantum Few-Body Problem.

### BSc and MSc in Physical Chemistry
**University of Pavia and University of Cambridge**

Sept 2009 – Sept 2014 | Pavia, Italy – Cambridge UK

MSc : A classical dynamic that conserves the quantum Boltzmann distribution.

BSc: Molecular Memories: rotaxane- and catenane-based molecular devices.

Second level MSc at Scuola Universitaria Superiore (IUSS)

## MOST PROUD OF

**Swiss National Science Foundation (SNSF) Fellowship**
Research Grant, October 2019 - Nov 2021

**Project Manager - from Jul 2023**
Define scope, plan and organize the execution of the POC

## TECH SKILLS

**C++20 with libraries Armadillo, Boost, Eigen, and API OpenMP 4.0, MPI 3.1, SIMD, HPC** ●●●●●

**C# 12, Q#, .NET Core, Nuget, RestAPI, WPF, Razor, Blazor, ML.NET, DI, MVC, MVVM, Docker, CI/CD** ●●●●●

**Eclipse, Visual Studio, ReSharper, Git, Mathematica, LaTex** ●●●●●

**Javascript, Python, CSS, AJAX, JSON CSS, HTML5, VBA, OpenACC, CUDA** ●●●●○

## STRENGTHS

Algorithm Optimization | Strong Math Foundation | Concurrent Computation | Distributed Computing | Quantum Computing | Software Design | Scrum and agile framework | Cloud Architectures

Data Modelling | Digital Business Process | UX | Actuarial Methods | Accounting Standards IFRS17

Fast Learning | Scientific Writing | Teaching | Public Speaking | Cross Domain Knowledge

## LANGUAGES

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|---|---|
| German | A2 |
| English | Proficient |
| Italian | Mother Tongue |

## ABOUT ME

## PUBLICATIONS SELECTION OF

J. Chem. Theory Comput., 18, 234 (2022)
Quantum Proton Effects from Density Matrix Renormalization Group Calculations

Phys. Rev. A, 102, 022803 (2020)
Analytically projected, rotationally symmetric, explicitly correlated Gaussians with one-axis shifted centers

J. Chem. Phys., 142, 134103 (2015)
Boltzmann-conserving classical dynamics in quantum time-correlation functions: “Matsubara dynamics”